Course Name: Energy, Environment and Life Science
Course Code: EN-101

Contact Hours/Week: 2L **Course Credits:** 02

Course Objectives

- To acquire a basic understanding and knowledge about the environment and its allied problems.
- To acquire a basic understanding and knowledge of environmental policies.
- To understand the relationship between energy and environment.
- Realize the importance of ecosystem and biodiversity for maintaining ecological balance.
- Develop the ability to evaluate measures for the improvement and protection of environment.
- To develop analytical skills, critical thinking, and demonstrate problem-solving skills using scientific techniques towards solutions of current problems and prevention of future problems.

Unit Number	Course Content	Contact Hours
UNIT-01	**Life and its Diversity:** Classification and Evolution, Ecology: Organisms and Populations (Abiotic Factors, Responses to Abiotic Factors, Adaptations, Population Characteristics, Population Interactions like Mutualism Competition, Predation, Parasitism etc.); Ecosystem and its significance, Biodiversity and Conservation(Species on Earth and India, importance of Species Diversity, Loss of Biodiversity, In-situ and ex-situ conservation) ; Life processes (photosynthesis, Nutrition, Respiration, Transportation); Biology and Human Welfare (Human Health and Disease, Strategies for Enhancement in Food Production, Microbes like Algae, bacteria etc. in Human Welfare.	08L
UNIT-02	**Environment and its Resources:** Environmental Concerns in Global Politics, The Protection of Global Commons, Common Property Resources, India's Stand on Environmental Issues, Environmental Movements, Resource Geopolitics.	04L
UNIT-03	**Environmental Issues:** Air Pollution and Its Control, Water Pollution and Its Control, Solid Wastes, Agro-chemicals and their Effects, Radioactive Wastes, Greenhouse Effect and Global Warming, Ozone Depletion in the Stratosphere, Degradation by Improper Resource Utilization and Maintenance, Deforestation), Carbon footprint, International/National legal Environmental protection agencies.	04L
UNIT-04	**Energy and environment:** Conventional Energy sources for Power, Transportation and Heat generation, Renewable Energy resources alternatives, Carbon trading, Climate change and Sustainability concept, Technological Advances and Institutions in India for Renewable Energy programs, Energy and Environment policies	08L

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: Understand environmental problems arising due to developmental activities.
CO2: Identify the natural resources and suitable methods for conservation and sustainable development.
CO3: Realize the importance of ecosystem and biodiversity for maintaining ecological balance.
CO4: Identify the environmental pollutants and abatement devices.
CO5: Understand the national and international environmental issues.

Text Books:

1. Joseph B, Environmental Studies, 3rd Addition, McGraw Hill Education (India) Private Limited, 2018.
2. De AK, Environmental Chemistry, 7th Addition, New Age International (P) Limited, 2010.
3. NCERT, Environmental Science, New Delhi.

Reference Books:

1. Kaushik A, Kaushik CP, Environmental Science, New Age International (P) Limited, 2011.
2. Bharucha R, Environmental Studies, 2004.
3. Dave D, Katewa SS, Environmental Studies, Cengage Learning India Pvt. Ltd., 2012.